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FEDERAL SPECIFICATION

ALUMINUM ALLOY, 7011 OR 7008 ALCLAD 7079, PLATE AND SHEET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for use of all Federal Agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-250.)

1. SCOPE AND CLASSIFICATION

* 1.1 Scope. This specification covers the specific requirements for aluminum alloy, 7011 or 7008 alclad 7079, plate and sheet. The general requirements are covered in QQ-A-250. The plate and sheet covered by this specification shall be an integral composite product consisting of a heat-treatable aluminum alloy (7079) core with thin layers of an aluminum alloy (7011 or 7008) anodic to the core and of approximately equal thickness, bonded to both surfaces.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be as O, T6, T62, T651, or F temper, as specified (see 6.2). The definitions of these tempers, in accordance with ANSI H35.1, are as follows:

O	- Annealed.
T6	- Solution heat-treated and artificially aged.
T62	- Solution heat-treated from O or F temper to demonstrate response to heat-treatment, and artificially aged.
T651	- Solution heat treated, stress relieved by stretching to produce a permanent set of 2 percent nominal but not less than 1-1/2 nor more than 3 percent, and artificially aged. Plate shall receive no further straightening after stretching (see 6.3).
F	- As fabricated.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

FSC 9535

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Federal Specifications:

QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet, General Specification for

Federal Standards:

Fed. Std. No. 184 - Identification Marking of Aluminum, Magnesium and Titanium

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-I-8950 - Inspection, Ultrasonic,Wrought Metals, Process for

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute Publication

ANSI H35.1 - Alloy and Temper Designation System for Aluminum

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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3. REQUIREMENTS

3.1 Cladding material. Unless otherwise specified (see 6.2), material shall be supplied with either cladding alloy.

3.2 Chemical composition. The chemical composition of the core ingots or slabs and of the cladding plates used for the manufacture of the clad plates and sheets shall conform to the requirements specified in table I for core and cladding, respectively.

TABLE I - Chemical composition 1/

Element	Composition					
	7079 Core Alloy - Percent		7011 Cladding Alloy - Percent		7008 Cladding Alloy - Percent	
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Zinc	3.8	4.8	4.0	5.5	4.5	5.5
Magnesium	2.9	3.7	1.0	1.6	0.7	1.4
Copper	0.40	0.8	--	0.05	--	0.05
Chromium	.10	.25	0.05	.20	0.12	.25
Manganese	.10	.30	.10	.30	--	.05
Iron	--	.40	--	.20	--	.10
Silicon	--	.30	--	.15	--	.10
Titanium	--	.10	--	.05	--	.05
Others, each	--	.05	--	.05	--	.05
Others, total	--	.15	--	.15	--	.10
Aluminum	Remainder		Remainder		Remainder	

1/ Analysis shall regularly be made only for the elements specifically mentioned in the table. If,however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

3.3 Mechanical properties.

3.3.1 The mechanical properties of materials as supplied. The mechanical properties perpendicular to the direction of final rolling, except for material less than 9 inches in width, shall conform to the requirements of table II for the temper specified. For material less than 9 inches in width, the mechanical properties parallel to the direction of final rolling shall conform to the requirements of table II for the temper specified.

3.3.2 Mechanical properties, after heat treatment. In addition to conforming to the requirements of 3.3.1,material in the annealed (O) and the as fabricated (F) tempers shall, after proper solution heat-treatment and artificial aging, also conform to the requirements of table II for the T62 temper. Material as received in the T6 and T651 tempers shall, after proper resolution heat treatment and aging, be capable of conforming to the requirements of table II for the T62 temper.

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3.3.3 Bend test. Bend-test specimens taken from plate and sheet shall be capable of withstanding, without cracking, the bend test specified in QQ-A-250. The values for bend factor N are given in table III.

TABLE II - Mechanical properties

Temper	Thickness in inches	Tensile Strength, p.s.i. minimum	Yield Strength at 0.2 percent offset, p.s.i. minimum	Elongation in 2-in. or 4 times D, percent, minimum 1/ 2/
O	0.015 thru 0.499	40,000 3/	21,000 3/	10
	.500 thru 2.000	40,000 3/ 4/	--	10
T6	0.015 thru 0.039	69,000	60,000	7
	.040 thru .249	72,000	63,000	8
T651	0.250 thru 0.499	73,000	64,000	8
	.500 thru 1.000	74,000 4/	65,000 4/	8
T62 5/	0.015 thru 0.039	68,000	59,000	7
	.040 thru .062	69,000	60,000	8
	.063 thru .187	70,000	61,000	8
	.188 thru .249	71,000	61,000	8
	.250 thru .499	72,000	62,000	8
	.500 thru 1.000	73,000 4/	62,000 4/	8
F	All	6/	6/	6/

1/ Not required for material 1/2-inch or less in width.
2/ D represents specimen diameter.
3/ Maximum.
4/ The properties for these thicknesses are those of the core alloy since the tests are made on a round specimen machined from the plate.
5/ Material in the T62 temper is a user heat condition and is not available from the material producer.
6/ No requirements.

TABLE III - Bend test factor "N"

Thickness of material Inch	Tempers	
	O (annealed)	T6 and T62, (heat-treated and aged)
0.015 thru 0.020	1	7
.021 thru 0.062	2	8
.063 thru 0.091	3	9
.092 thru 0.125	4	10
.126 thru 0.249	5	11
.250 thru 0.499	6	14

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3.4 Cladding thickness.

3.4.1 Thickness of cladding plates. The aluminum alloy (7011 or 7008) plates that are bonded to the two sides of the aluminum alloy (7079) ingot or slab to form a composite that is to be rolled to the finished thickness shall each have a thickness not less than that specified in table IV.

TABLE IV - Cladding thickness

Thickness of finished plate or sheet in inches	Nominal cladding thickness per side; percent of composite thickness	Average thickness per side of cladding on finished plate or sheet; minimum percent of plate or sheet thickness
Under 0.063	4.0	3.2
0.063 thru 0.187	2.5	2.0
0.188 and over 1/	1.5	1.2

1/ For plate 0.500-inch and over, the average cladding thickness per side shall have a maximum value of 3 percent of the plate thickness.

3.4.2 Average thickness of cladding. When specified (see 6.2), the average cladding thickness of the finished sheet or plate shall be determined. Samples, examined in accordance with QQ-A-250, shall show an average of cladding on each side as specified in table IV.

3.5 Internal defects. When specified (see 6.2), plate shall be ultrasonically inspected (see QQ-A-250). Acceptance limits shall be as specified in table V.

TABLE V - Ultrasonic discontinuity acceptance limits 1/

Thickness in inches	Maximum weight per piece in pounds	Discontinuity class 2/
0.500 thru 1.499	2,000	B
1.500 thru 2.000	2,000	A

1/ Discontinuities in excess of those listed in table V may be allowed subject to the approval of the procuring activity, if it is established that they will be removed machining or that they are in non-critical areas.

2/ The discontinuity class limits are defined in MIL-I-8950.

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3.6 Marking. In addition to the marking required in Fed. Std. No. 184, plate and sheet in the T6 and T651 tempers shall be identified by a lot number marked in at least one location and each piece.

4. QUALITY ASSURANCE PROVISIONS (See QQ-A-250).

4.1 Mechanical tests after heat treatment.

4.1.1 Number of tests after heat treatment. From material in the annealed (O) and as-fabricated (F) tempers, an additional number of specimens equal to that required by QQ-A-250 shall be taken and tested after solution heat treatment and artificial aging to determine compliance with 3.3.2.

5. PREPARATION FOR DELIVERY (See QQ-A-250).

6. NOTES

6.1 Intended use. 7011 or 7008 Alclad 7079 plates and sheets are used where fatigue strength and mechanical properties higher than those provided by 7072 Alclad 7079 are required (see MIL-A-8923).

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Form and temper required (see 1.2.1).
c. Dimensions required.
d. Requirements for sizes not specifically covered (see QQ-A-250).
e. Whether 7011 or 7008 cladding is to be specified (see 3.1).
f. Whether cladding thickness determination is required (see 3.4.2).
g. Whether ultrasonic inspection of material is required (see 3.5).
h. Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-250).

6.3 Temper. Temper T651 is available in plate only.

6.4 Definitions.

6.4.1 Capable of. The term "capable of" as used in the detailed specification means that the test need not be performed by the producer of the plate and sheet. However, should subsequent testing by the procuring activity establish that the material does not meet these requirements, the plate and sheet are subject to rejection.

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QQ-A-250/27A ALUMINUM ALLOY, 7011 OR 7008 ALCLAD 7079, PLATE AND SHEET

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MATERIAL PROCURED UNDER A

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